8/8/2026

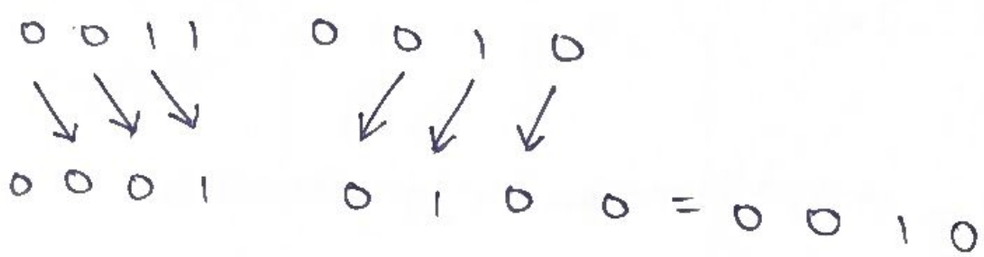
Computer Architecture

Assignment - 1

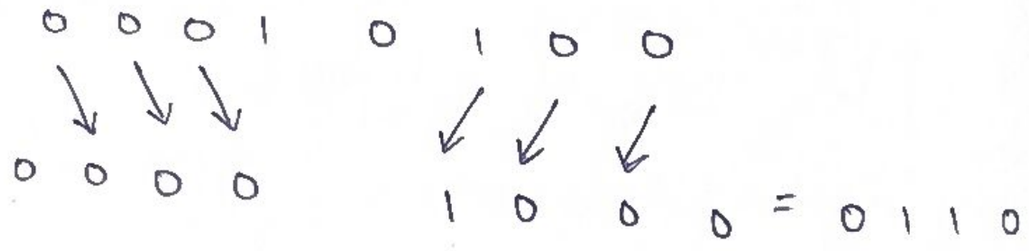
1) 3*2 multiplication in binary multiplicatio

Sol:- Steps q_0 q_1

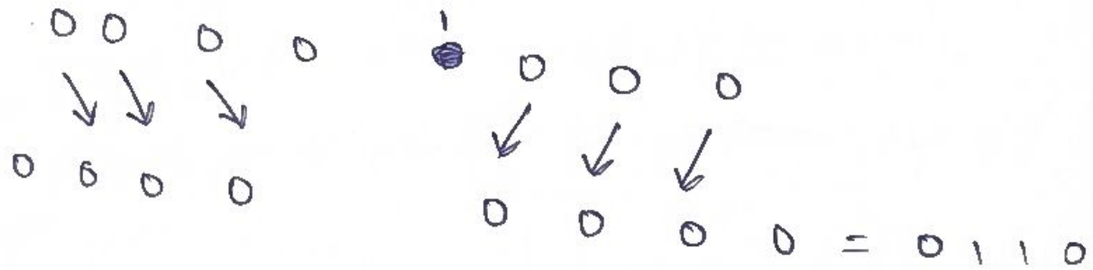
1.



2.



3.



$3*2 = 0\ 0\ 1\ 1 \times 0\ 0\ 1\ 0$

$= 1\ 0\ 1\ 1\ 0$

$\Rightarrow 6$

2) $-6 + 2$ 1's complement and 2's complement for binary addition.

Sol:- $-6 = 0\ 0\ 1\ 1\ 0$

∴ Taking the 1's complement for (-6)

0 0 1 1 0 ⇒ 1 1 0 0 1

∴ Taking 2's complement

```
  1 1 0 0 1
        1
  ---------
  1 1 0 0 0   ⇒ 2's complement
```

```
          (1)
-6 + 2 = 1 1 0 1 0
         0 0 0 1 0
         ---------
         1 1 1 0 0
         ---------
```

Taking 1's Complement of -6 + 2

1 1 1 0 0 = 0 0 0 1 1

Taking 2's Complement of -6 + 2

```
        1 1
  0 0 0 1 1
          1
  ---------
  0 0 1 0 0
  ---------
```

-6 + 2 = 11010 + 00010

⇒ **00100** = -4